- The Ionization Of Water Is Expressed By an Equilibrium Constant

- The degree of ionization of water at equilibrium is small
- At 25^{0}C only about two out of every 10^9 molecules in pure water are ionized at any instant

- The equilibrium constant for the reversible ionization of water is

$$K_{eq} = \frac{[H^+][OH^-]}{[H_2O]}$$

According to the **law of mass action** the rate of a chemical reaction is proportional to the product of the masses of the reactants. Necessarily, this implies that for a chemical reaction mixture that is in equilibrium, the ratio between the concentration of reactants and products is constant

$$K_{eq} = \frac{[H^+][OH^-]}{[H_2O]}$$

- Since 1 mole (mol) of water weighs 18 g,
- 1 liter (L) (1000 g) of water contains
- 1000 ÷ 18 = 55.56 mol
- Pure water thus is 55.56 molar(M)
- Accordingly, we can substitute 55.5 M in the equilibrium constant expression to yield
- Keq = $[H^+][OH^-]/55.5$
- molecular wt. of water is 18.01528 g/mol

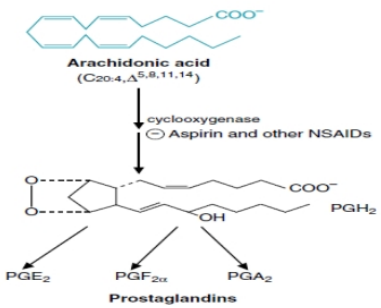
- Keq = $[H^+][OH^-]/55.5$
- On rearranging, this becomes

INHIBITION OF ENZYME ACTIVITY

- Enzyme inhibitors are molecular agents that interfere with catalysis, slowing or halting enzymatic reactions.
- Any substance that can diminish the velocity of an enzyme-catalyzed reaction is called an inhibitor.

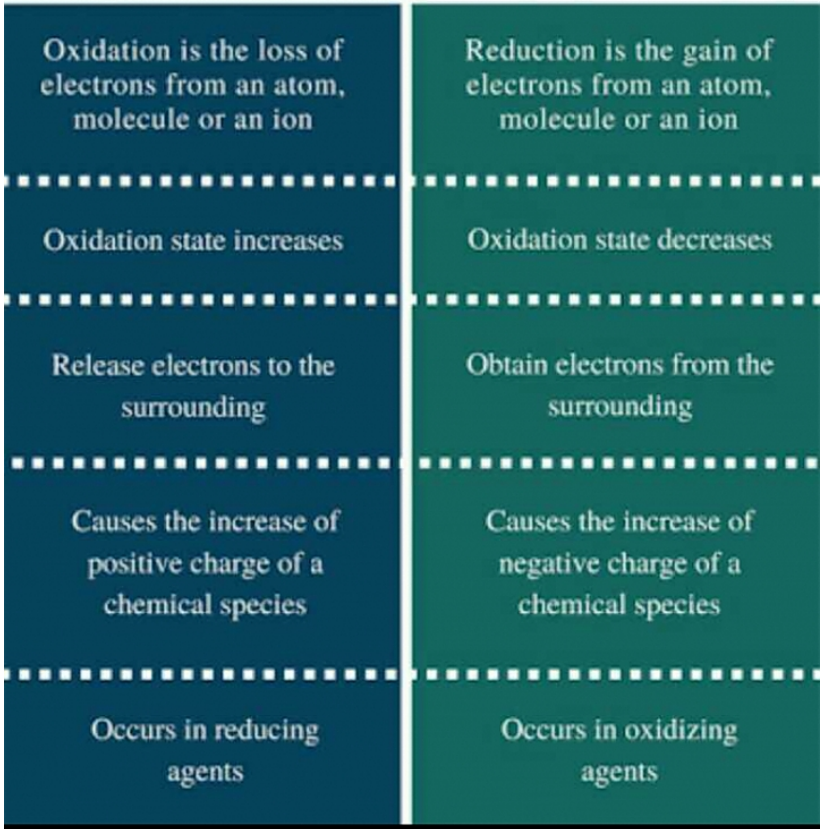
- Enzymes catalyze virtually all cellular processes, so it should not be surprising that enzyme inhibitors are among the most important pharmaceutical agents known.

For example, aspirin inhibits the enzyme (cyclooxygenase) that catalyzes the first step in the synthesis of prostaglandins, compounds involved in many processes, including some that produce pain.



- Inhibitors can be classified on the basis of their site of action on the

OXIDATION VERSUS REDUCTION



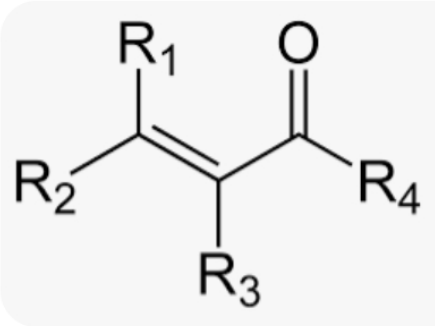
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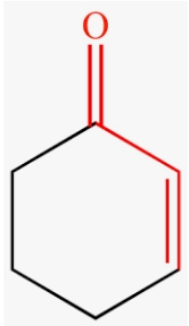
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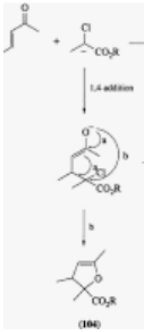
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enone

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An enone is **a type of an α,β-unsaturated carbonyl that consists of an alkene conjugated to a ketone**. The simplest enone is methyl vinyl ketone (butenone, CH_2=$CHCOCH_3$). Enones are typically produced using an aldol condensation or Knoevenagel condensation.

https://en.m.wikipedia.org › wiki

Alpha-beta Unsaturated carbonyl

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- The difference between the energy levels of the ground state and the transition state is the **activation energy, *G‡*.**

- At the top of the energy hill is a point at which decay to the S or P state is equally probable (it is downhill either way)

- A substance that modifies the transition state to lower the activation energy is termed a catalyst; a biological catalyst is termed an enzyme.

END

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- But there is an **energy barrier** between S and P: The energy required for
- alignment of reacting groups
- formation of transient unstable charges
- bond rearrangements
- and other transformations

- This is illustrated by the energy "hill"
- To undergo reaction, the molecules must overcome this barrier and therefore must be raised to a higher energy level
- This is called the **transition state**.

- It is simply a fleeting molecular moment in which events such as
- bond breakage,
- bond formation, and
- charge development
- have proceeded to the precise point at which decay to either

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All animal and plant cells are powered by chemical energy stored in the chemical
Bonds of organic molecules. **Animals convert the chemical energy of fuels into motion and heat and**

In a few organisms such as fireflies chemical energy is converted into light…

Lets look at some Examples of major bioenergetic processes occurring in nature
ATP is the energy currency in biological systems
Citric acid cycle, that is the process of cellular respiration generates molecules of ATP by oxidation of energy stored in food material
These processes will be discussed in detail in coming lectures.

Laws of Thermodynamics

- There are two fundamental laws of thermodynamics.

1. First law

The first law is the principle of the conservation of energy: it states that

"for any physical or chemical change, the total amount of energy in the universe remains constant; energy may change form or it may be transported from one region to another, but it cannot be created or destroyed."

Entropy

- The term "entropy," was first used in 1851 by Rudolf Clausius, which literally means "a change within,"

Entropy = disorder

- **Entropy** is a quantitative expression for the randomness or disorder in a system. When the products of a reaction are less complex and more disordered than the reactants, the reaction is said to proceed with a gain in entropy.
- Denoted by **S**
- ΔS has a +ve sign when entropy increase and –ve when entropy decreases
    - units of entropy are joules/mole.Kelvin (J/mol.K) (joules/mole **multiply by** Kelvin)

The term "entropy," was first used in 1851 by Rudolf Clausius, which literally means "a change within" a system.. and it can be described as disorder..
By convention, ΔS has a positive sign when entropy increases and ΔH, as noted above, has a negative sign when heat is released by the system to its surroundings.

Second Law of Thermodynamics

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Select text

- Inhibitors can be classified on the basis of their site of action on the enzyme,
- on whether they chemically modify the enzyme, or on the kinetic parameters they influence.
- Two broad classes of enzyme inhibitors:
- **1. Reversible**
- **2. Irreversible.**

- In general, irreversible inhibitors bind to enzymes through covalent bonds.
- Reversible inhibitors typically bind to enzymes through noncovalent bonds,

dilution of the enzyme–inhibitor complex results in dissociation of the **reversibly**
bound inhibitor, and recovery of enzyme activity.

- The two most commonly encountered types of reve inhibition are;
- competitive and
- noncompetitive.

BIOENERGETICS

TYPES OF CHEMICAL REACTIONS IN CELLS

- **Chemical Logic and Common Biochemical Reactions**
- Most cells have the capacity to carry out thousands of specific, enzyme-catalyzed reactions:
- for example, transformation of a simple nutrient such as glucose into amino acids, nucleotides, or lipids
- extraction of energy from fuels by oxidation
- polymerization of monomeric subunits into macromolecules

5 types of chemical reactions in cells

The reactions going on in biological systems can be categorized into 5 types:

1. Reactions that make or break carbon-carbon bonds
2. Internal rearrangements, isomerizations, and eliminations
3. Free-radical reactions
4. Group transfer
5. Oxidation-reduction reactions

2 basic chemical principles

(1) A covalent bond can be broken in two general ways

- Homolytic cleavage
- Heterolytic cleavage

- A **Covalent bond**, also called a molecular *bond*, involves the sharing of electron pairs between atoms.
- These electron pairs are known as shared pairs or *bonding* pairs, and

Water

- Water is the most abundant substance in living systems, making up 70% or more of the weight of most organisms

- The hydrogen bonding between water molecules and the slight tendency of water to ionize are of crucial importance to the structure and function of biomolecules

Hydrogen Bonding gives Water its Unusual Properties

- Water has a higher
- melting point,
- boiling point, and
- heat of vaporization than most other common solvents

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inhibitor 1 of 44

INHIBITION OF ENZYME ACTIVITY

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The pH Scale

- Designates the H^+ and OH^- Concentrations
- The pH of a solution is defined as the logarithm to the base 10 of the reciprocal of the $[H^+]$, i. e, the negative logarithm of the $[H^+]$
- pH = log $1/[H^+]$
- = $-\log[H^+]$

The symbol **p** denotes "negative logarithm of"

- For each pH unit less than 7.0, the $[H^+]$ is increased tenfold;
- for each pH unit above 7.0, it is decreased tenfold

- The pH of water at 25°C, in which H^+ and OH^- ions are present in equal numbers, is 7.0
- pH = $-\log[1 \times 10^{-7}] = 7$

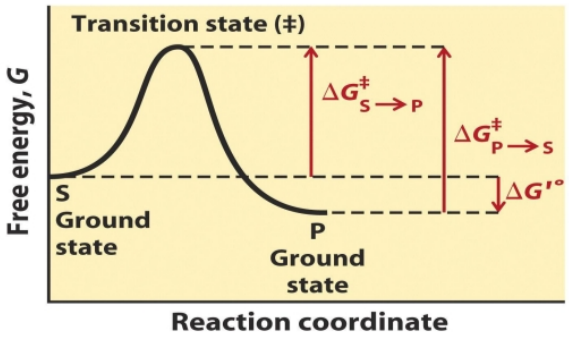
- The symbol p denotes "negative logarithm of"

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- Formic acid and not CO_2 is the primary reaction product of phytanic acid oxidation. The amount of formic acid was shown to be about 9-fold higher as compared to CO_2

- **Origin of Phytanic Acid**
- Phytol is the alcohol moiety of the chlorophyll molecule and, as an integral part of chlorophyll, it is abundantly present in plants and even in the marine environment
- Phytol can be released from chlorophyll by the action of bacteria present in the rumen of ruminant animals.
- The resulting free **phytol** can be converted to phytanic acid in the rumen of animals





Activation energies, *G‡,*

- The activation energies, *G‡, for* the
- **S → P and P → S** reactions are indicated.
- *G'° is the overall standard* free-energy change in the direction **S → P**

Double dagger . Δ *G'*.

Reaction coordinate diagram for a chemical reaction.
The free energy of the system is plotted against the progress of the reaction S→P. A diagram of this kind is a description of the energy changes during the reaction, and the horizontal axis (reaction coordinate) reflects the progressive chemical changes (e.g., bond breakage or formation) as S is converted to P. The activation energies, *G‡*, for the S→P and P→S reactions are indicated. *G* is the overall standard free-energy change in the direction S→P.

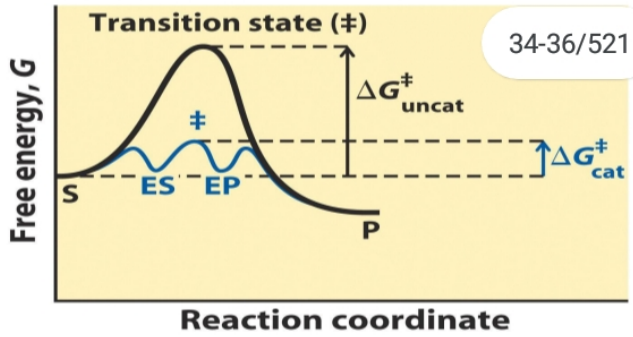
change for the reaction, ΔG.

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Standard free-energy change, ΔG^{o}

- Under standard conditions (298 K = 25 C)
- when reactants and products are initially present at 1 M concentrations or
- for gases, at partial pressures of 101.3 (kPa), or 1 atm





Reaction coordinate diagram comparing enzyme catalyzed and uncatalyzed reactions. In the reaction S→P, the ES and EP intermediates occupy minima in the energy progress curve of the enzyme-catalyzed reaction. The terms $G^{\ddagger}_{uncat}$ and $G^{\ddagger}_{cat}$ correspond to the activation energy for the uncatalyzed reaction and the overall activation energy for the catalyzed reaction, respectively. The activation energy is lower when the enzyme catalyzes the reaction.

- The role of enzymes is to accelerate the inter-conversion of S and P.
- i.e ***enzymes lower the energy of activation, $\Delta G^{\ddagger}$, of a reaction.***
- The enzyme is not used up in the

- The role of enzymes is to accelerate the inter-conversion of S and P.
- i.e ***enzymes lower the energy of activation, $\Delta G^{\ddagger}$, of a reaction.***
- The enzyme is not used up in the process, and the equilibrium point is unaffected.

- However, the reaction reaches equilibrium much faster when the appropriate enzyme is present,
- because the rate of the reaction is increased.

END

The Induced Fit Hypothesis

- Some proteins can change their shape (conformation)
- When a substrate combines with an enzyme, it induces a change in the enzyme's conformation

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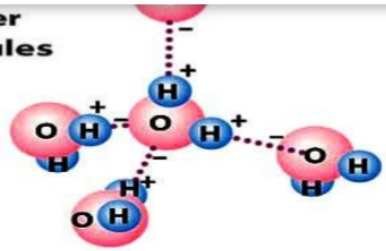
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Second Law of Thermodynamics



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of water molecules



Ionization of Water

- Although many of the solvent properties of water can be explained in terms of the uncharged H_2O molecule, the small degree of ionization of water into hydrogen ions (H^+) and hydroxide ions (OH^-) must also be taken into account

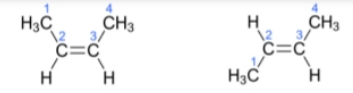
- Pure Water Is Slightly Ionized
- Water molecules have a slight tendency to undergo reversible ionization to yield a hydrogen ion (a proton) and a hydroxide ion, giving the equilibrium reaction

- $H_2O \leftrightarrow H^+ + OH^-$

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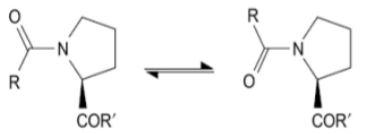
(b) Cis-trans rearrangement

- In general, Peptide bonds formed between amino acid residues with alpha carbons in trans-configuration... (more than 99.95% have **trans configuration**.)
-
- For peptide bonds involving the **imino nitrogen of proline, peptide bonds** are able to exist both in **cis and trans configuration.. (6% are in the cis configuration)** through peptidyl-prolyl (proline) isomerase enzyme



cis-butene

trans-butene



cis *trans*

The *cis* and *trans* isomers of proline.
The process of the formation of *trans* and *cis* conformation is via isom
and the reaction is a reversible process.

Proline

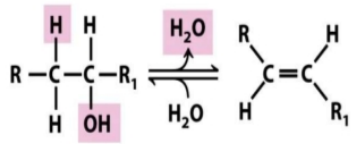
- **Cis–trans** isomerism, also known as geometric isomerism or configurational isomerism. Cis is illustrated by... The presence of functional groups on the same side of the carbon chain while **trans** conveys that functional groups are on opposing sides of the carbon chain.

- So while considering this kind of isomerism in living system, we can quote the example of proline amino acid. In general, Peptide bonds formed between amino acid residues are in trans-configuration... more than 99.95% have **However, in case of proline, Peptide bonds** formed are able to exist both as **cis and trans isomers..** through peptidyl-prolyl (proline) isomerase enzyme

Prolyl isomerase ..this enzyme is (EC 5.2.1.8) found in both prokaryotes and eukaryotes and interconverts the cis and trans isomers of peptide bonds with the amino acid proline.

(c) Elimination

- **The loss of water from an alcohol, resulting in the introduction of a C=C** bond b/w 2 adjacent carbon atoms
- **An elimination reaction with no changes in oxidation state**



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- A substance that modifies the transition state to lower the activation energy is termed a catalyst; a biological catalyst is termed an enzyme.



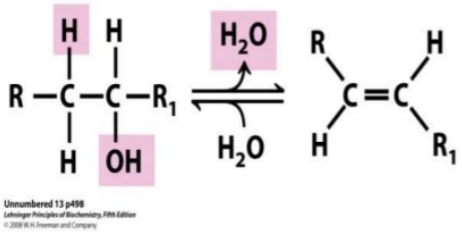
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Loss of water from an alcohol → introduction of a C=C bond

3) Free-radical reactions

• Homolytic cleavage of covalent bond → produce free radical

1) **Isomerizations** that make use of adenosylcobalamin (vitamin methylmalonyl-CoA mutase (MCM) reactions